How fast is **India** catching up?

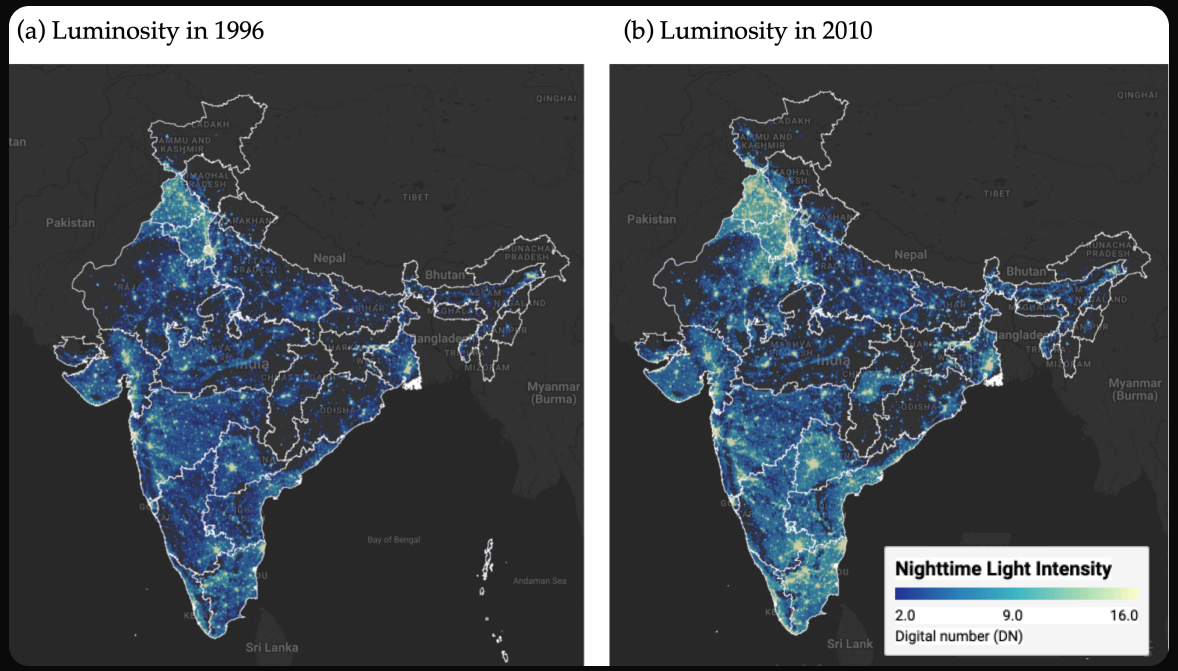


Figure 1. Original map and color scale, CC BY 4.0

Regional growth, convergence, and spatial spillovers in India: A reproducible view from outer space

520 districts, one growth interval

520

Indian districts

1996–2010

Study period

Radiance-calibrated DMSP-OLS

14

Years of growth

One cross-sectional growth interval

Nighttime lights per person serve as a proxy for economic activity.

Three **empirical extensions**

01

Visualize

Interactive satellite imagery

02

Test dependence

Spatial clustering and model diagnostics

03

Model spillovers

Spatial Durbin model

The paper extends Chanda and Kabiraj (2020) through a reproducible workflow.

Initial luminosity and growth **cluster**

0.73

Initial luminosity

Global Moran's I · p = 0.001

0.60

Luminosity growth

Global Moran's I · p = 0.001

Neighboring districts have correlated luminosity levels and growth rates.

A **spatial model** separates associations

−0.025

Direct effect

Within-district association

−0.012

Indirect effect

Association through neighbors

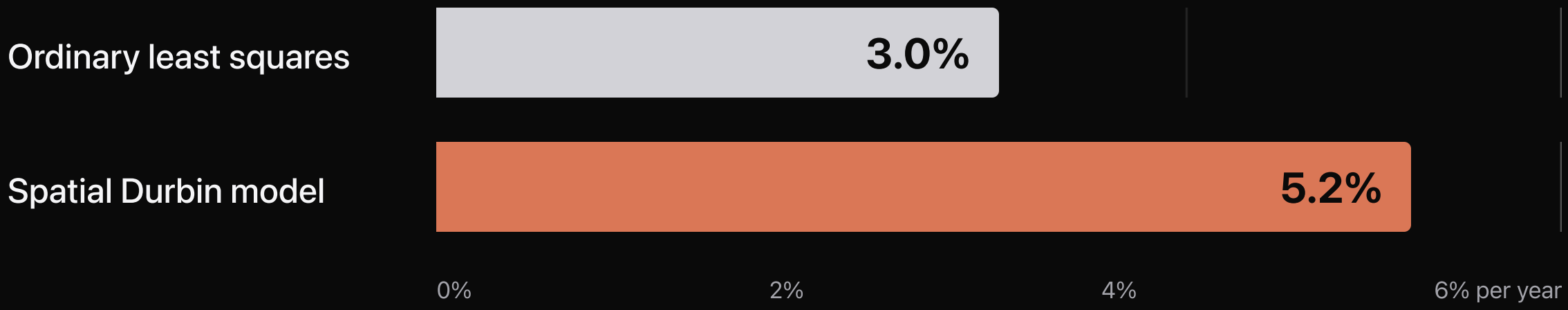
−0.037

Total effect

Direct plus indirect

These are estimated effects of initial log luminosity on subsequent growth in Model 4.

The estimated **convergence speed** rises



The preferred Model 4 includes control variables and state fixed effects.

The implied **half-life** falls

23

years

Ordinary least squares

13

years

Spatial Durbin model

Preferred Model 4

Half-life is the model-implied time to close half the initial gap to the steady state.

The pattern survives **alternative weights**

Spatial weights	Direct effect	Total effect
4 nearest neighbors	−0.024	−0.032
6 nearest neighbors	−0.025	−0.037
8 nearest neighbors	−0.025	−0.036
Queen contiguity	−0.025	−0.035
Rook contiguity	−0.025	−0.034
Inverse distance	−0.025	−0.041
Inverse squared distance	−0.025	−0.037

Direct and total convergence effects are negative and statistically significant in all seven specifications.

Interpretation remains **associational**

MODEL

Shared conditions

Geography, institutions, infrastructure and omitted variables can contribute to spatial dependence.

MEASUREMENT

A limited observation window

One growth interval and a luminosity proxy constrain what the estimates explain.

The analysis does not identify a causal effect of changing one district on another.

An exploratory link to **culture**

+0.370

Media-based consumption

Spearman correlation · p = 0.037

−0.404

Community participation

Spearman correlation · p = 0.022

Across 32 states and union territories, luminosity relates differently to these two forms of cultural participation.

Six notebooks make the evidence **inspectable**

N1

Earth Engine

Open notebook

N2

Convergence

Open notebook

N3

Spatial dependence

Open notebook

N4

Spillover modeling

Open notebook

N5

Robustness

Open notebook

N6

Spatial culture

Open notebook

N1 connects to Earth Engine, and N2–N6 provide analytical Python workflows.

Spatial connections change estimated **catch-up**

3.0%

OLS

versus

5.2%

SDM

Published paper Online article Code and data

Accounting for spatial dependence raises estimated convergence while leaving causal interpretation open.

REGION 13(2), 173–196 · CC BY 4.0